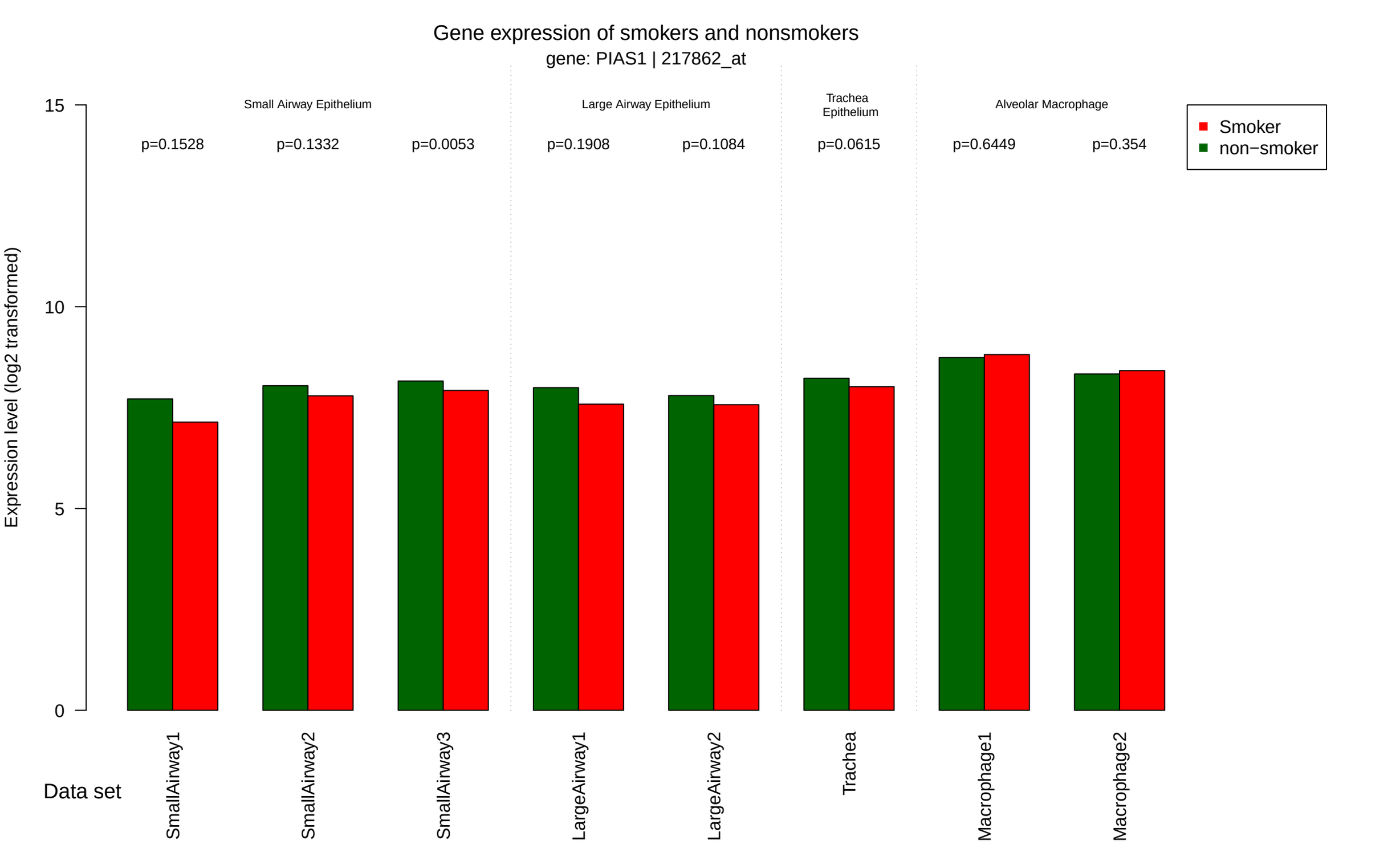
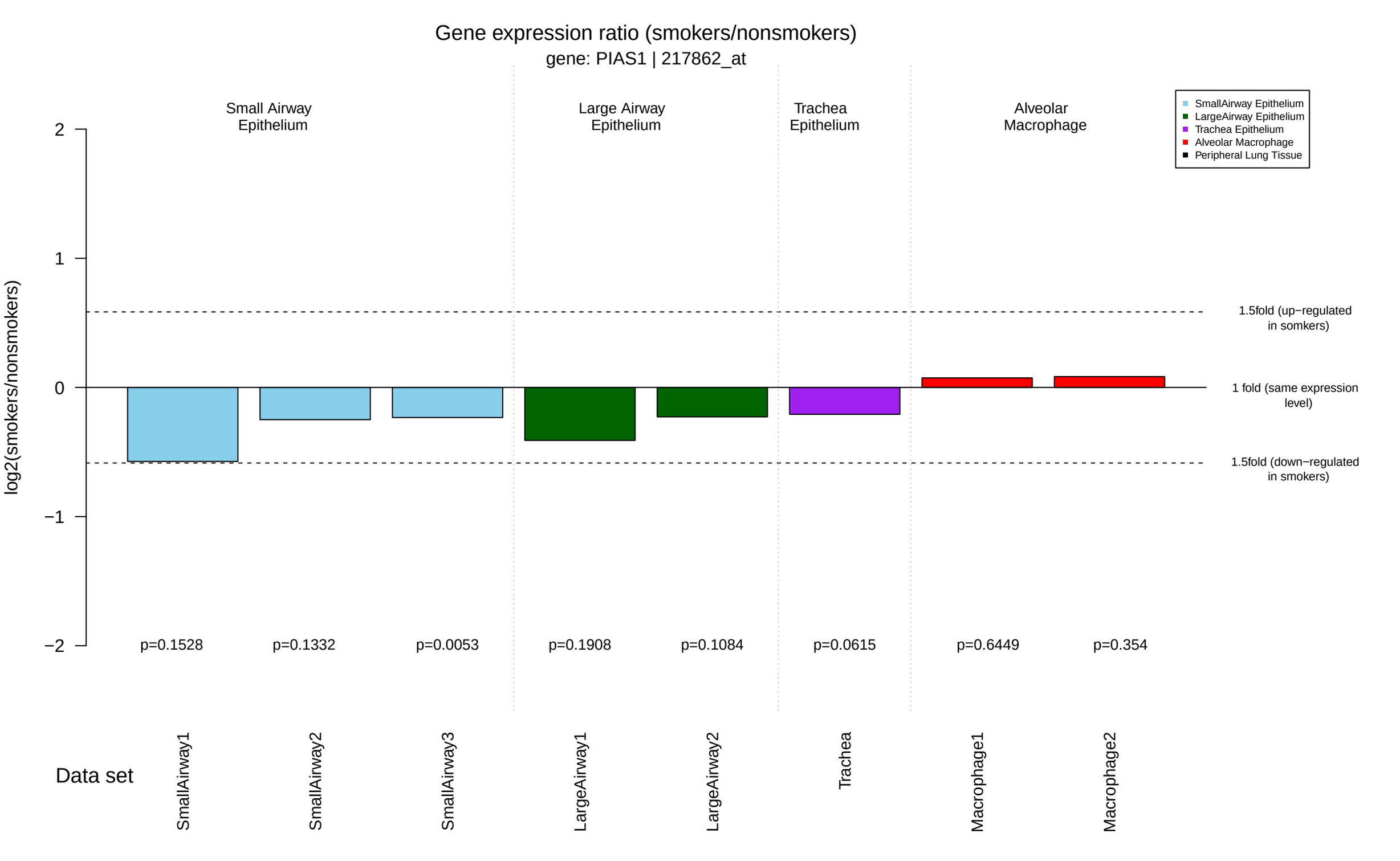
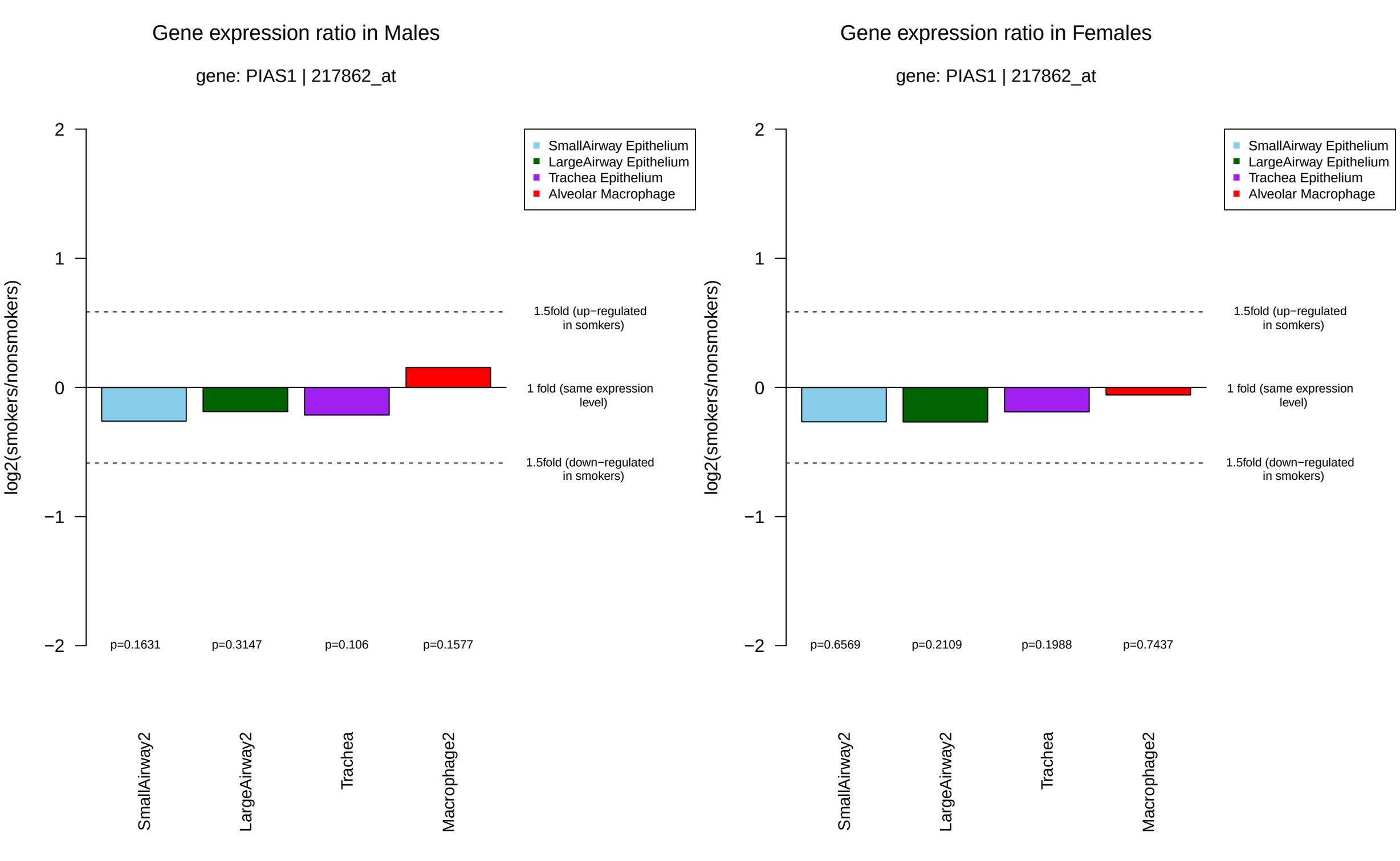
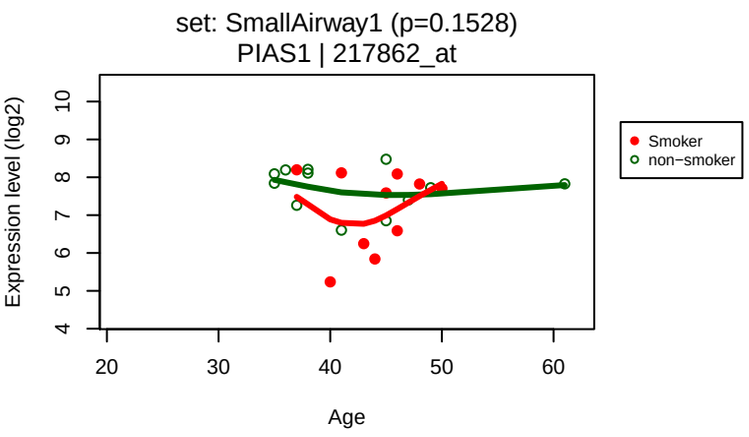
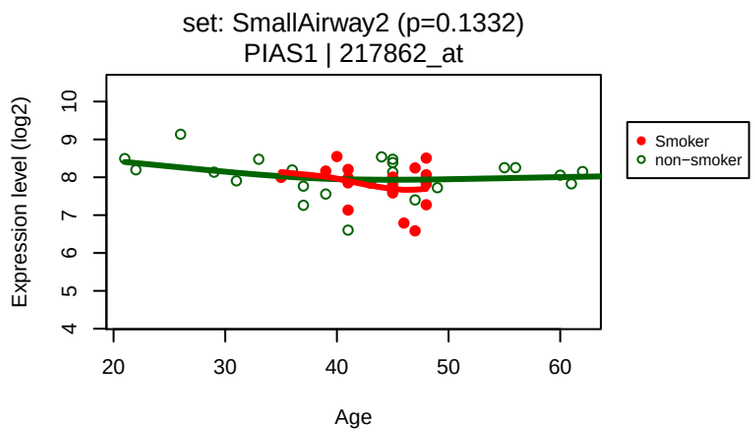
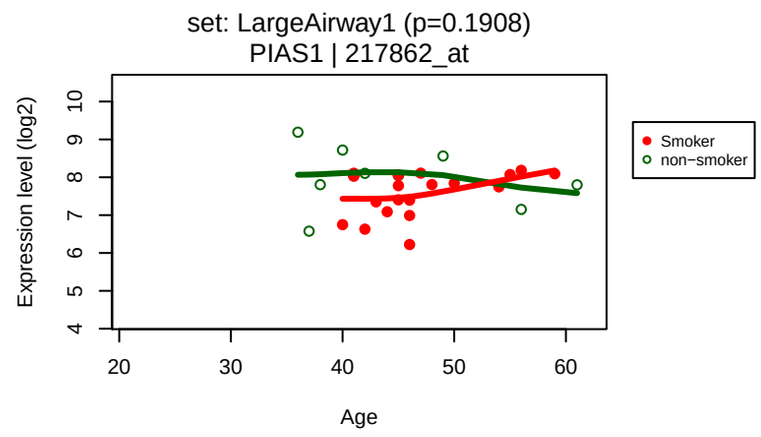

Gene expression of smokers and nonsmokers
gene: PIAS1 | 217862_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1528
p=0.1332
p=0.0053
p=0.1908
p=0.1084
p=0.0615
p=0.6449
p=0.354
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

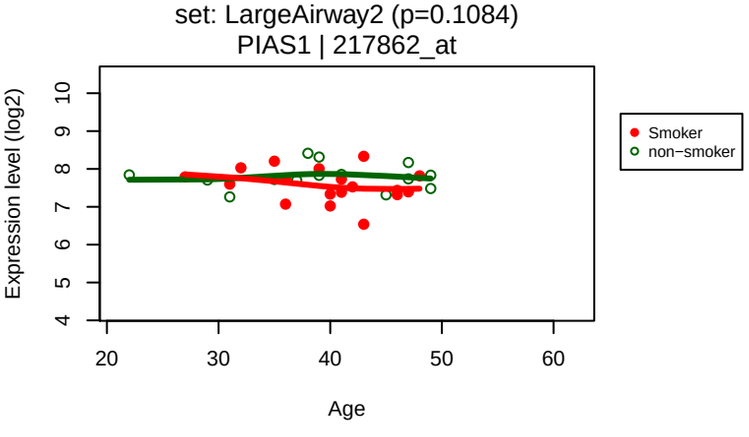

Gene expression ratio (smokers/nonsmokers)
gene: PIAS1 | 217862_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1528
p=0.1332
p=0.0053
p=0.1908
p=0.1084
p=0.0615
p=0.6449
p=0.354
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

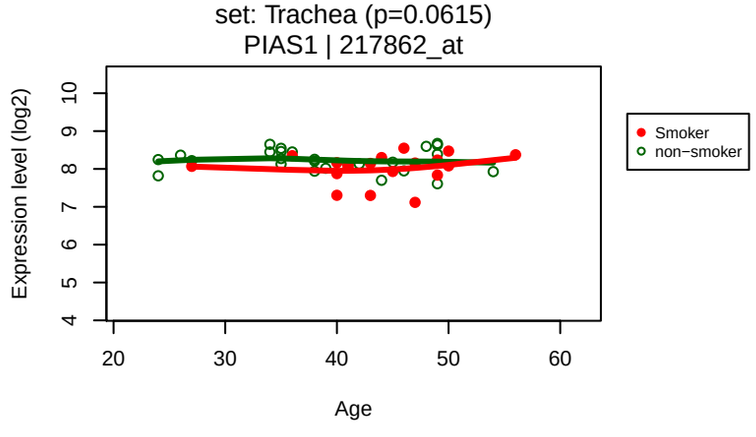

Gene expression ratio in Males
gene: PIAS1 | 217862_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1631
p=0.3147
p=0.106
p=0.1577
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIAS1 | 217862_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6569
p=0.2109
p=0.1988
p=0.7437
SmallAirway2
LargeAirway2
Trachea
Macrophage2

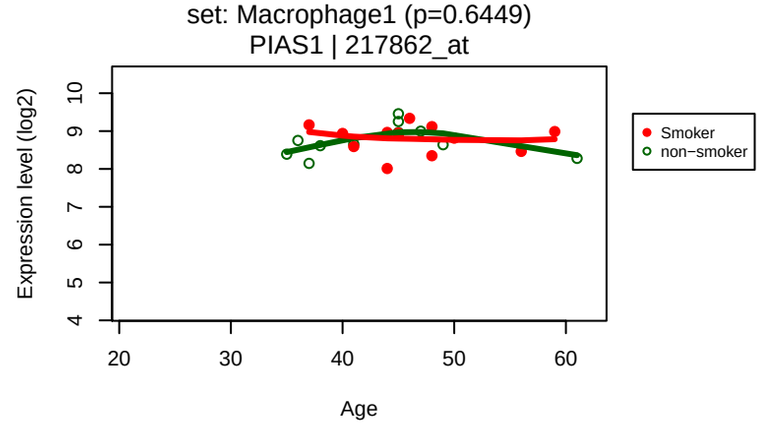

set: SmallAirway1 (p=0.1528)
PIAS1 | 217862_at
Expression level (log2)
Age
Smoker
non−smoker

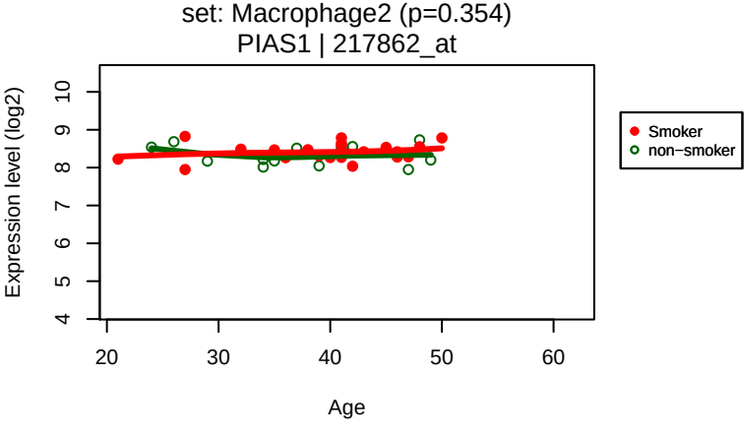

set: SmallAirway2 (p=0.1332)
PIAS1 | 217862_at
Expression level (log2)
Age
Smoker
non−smoker

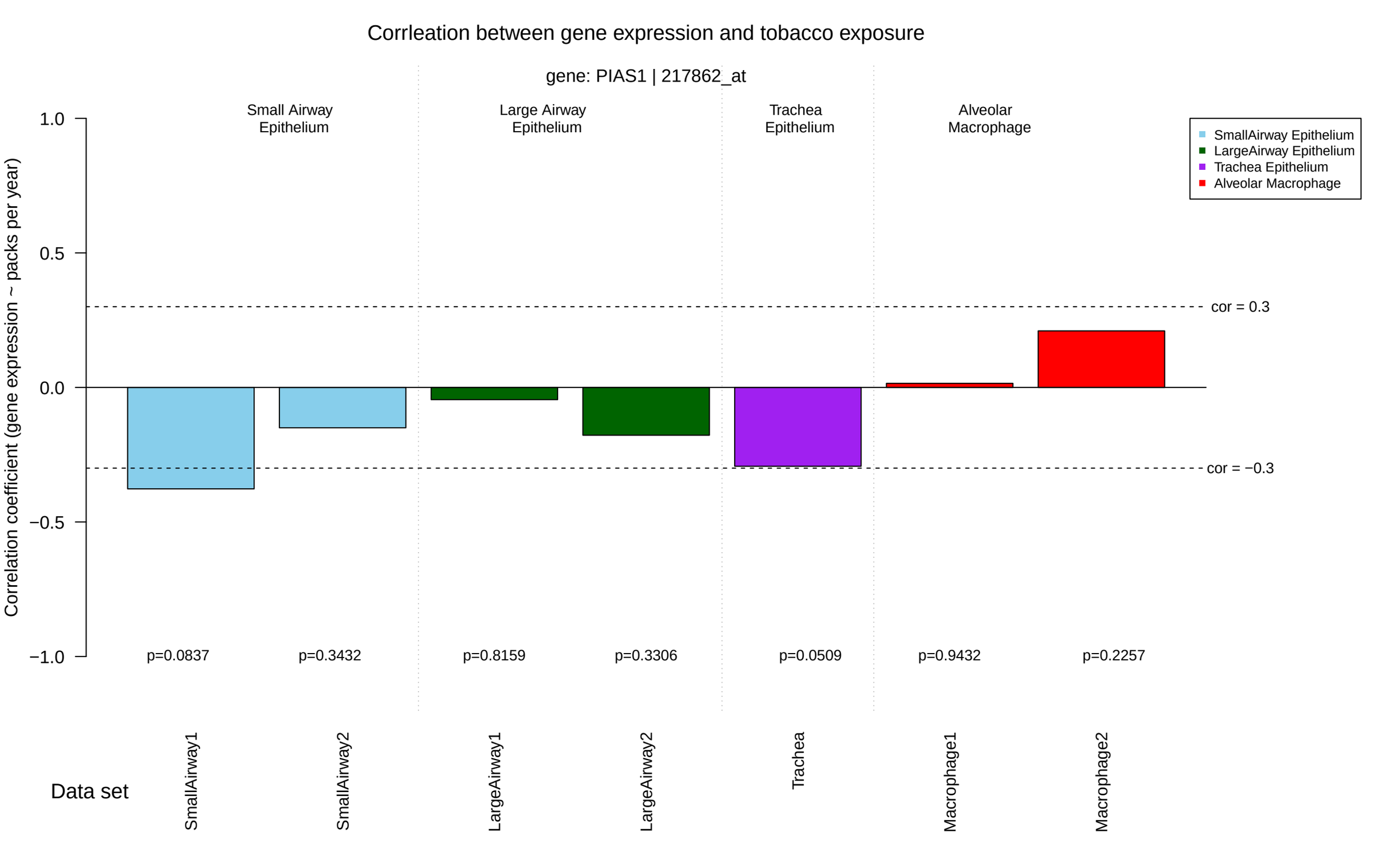

set: LargeAirway1 (p=0.1908)
PIAS1 | 217862_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1084)
PIAS1 | 217862_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0615)
PIAS1 | 217862_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6449)
PIAS1 | 217862_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.354)
PIAS1 | 217862_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIAS1 | 217862_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0837
p=0.3432
p=0.8159
p=0.3306
p=0.0509
p=0.9432
p=0.2257
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2